



DIGITAL TECHNOLOGY

Academic Year 2025-26

Prof. Barbara PERNICI

Exam 23/6/2026

Total time 1.30 h

Last name

First name

Matricola/Person code

Signature

Please remember that:

- The exam is closed-book.
- The use of cellular phones or any other electronic devices during the exam is forbidden.

It is necessary to answer at least partially all the questions for a positive evaluation.

Given answers should be explained; lists of bullet items are insufficient to answer a question.

Please write the answer for the Python exercise on a separate piece of paper.

Question 1 [11 points]

Define the data quality dimensions studied in the course.

For at least two data quality dimensions, describe the techniques used in the course to assess the quality of a dataset for these dimensions and to improve it.

Question 2 [11 points]

Illustrate the process to design data analysis dimensions in a data warehouse, discussing the models illustrated in the course.

[please turn over]

Question 3 [11 points]

Consider the following fragment of Python code:

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```
import pandas as pd

records = [('Alice', 320), ('Bob', 85), ('Alice', 410), ('Carol', 150), ('Bob',
230)]

orders = pd.DataFrame(records, columns=['name', 'amount'])
```

```
info = pd.DataFrame({
    'name':    ['Alice', 'Bob',    'David'],
    'city':    ['Rome',   'Milan', 'Naples'],
})

merged = orders.merge(info, on='name', how='left')
```

Questions:

- 1) Add a new column to `orders` containing the values in column 'amount' minus the mean of 'amount'.
- 2) What is the value of the attribute 'city' for any record corresponding to Carol in `merged`, and why?
- 3) Calculate the total amount per 'city'.